

# Evidence Summary - Technology #14

## Acoustic Obstruction Detection for CSF Shunts

### External Evidence Sources (5 total):

#### Source 1: Acoustically Active Catheter for Intracardiac Navigation by Color

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5515670>

Excerpt: Jun 5, 2017 · We present a catheter fitted with a piezoelectric crystal, which vibrates and produces an instantaneous marker in color flow Doppler scans.

#### Source 2: Acoustic Wave Catheter - Active Life Medical Products

URL: <https://activelifemed.com/glossary/acoustic-wave-catheter-learn-more-in-our-activelifemed-glossary>

Excerpt: Oct 16, 2024 · An acoustic wave catheter is a specialized medical device designed to deliver therapeutic ultrasound energy to targeted tissues within the body.

#### Source 3: Mechanisms of Focused Ultrasound

URL: <https://www.fusfoundation.org/the-technology/mechanisms-of-focused-ultrasound>

Excerpt: Neuromodulation with focused ultrasound allows noninvasive stimulation or suppression of neural activity. Low-intensity ultrasound can reversibly alter brain or ...

#### Source 4: A flexible catheter-based sensor array for upper airway soft tissues

URL: <https://www.nature.com/articles/s41467-024-55088-y>

Excerpt: Jan 2, 2025 · Here, we report a catheter-based flexible pressure sensor array that continuously monitors soft measuring upper airway obstruction location<sup>1,2</sup>. ...

#### Source 5: Advanced Urology Therapy Device In Tyler, TX | UroShield

URL: <https://nanovibronix.com/uroshield-product-information>

Excerpt: The ultrasonic waves generated by the clip create an acoustic shield on the surfaces of the catheter to interfere with the attachment of bacteria, prevent ...

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.  
No evidence has been physically validated.*